

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics

Extension 1

Year 11
PRELIMINARY HSC ASSESSMENT TASK 2
JULY 2007

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- Marks may not be awarded for careless or badly arranged work

NAME : _____

QUESTION 1	QUESTION 2	QUESTION 3	QUESTION 4	QUESTION 5	TOTAL

Question 1**Marks**

- a) Find the maximum value of $8 - 4x - x^2$ 1
- b) Solve for x : $x(2x - 9) < 11$ 2
- c) If A is acute and $\sin A = \frac{1}{3}$ find the exact value of
- i) $\cos (90^\circ - A)$ 1
- ii) $\sin 2A$ 2
- d) Evaluate $\lim_{x \rightarrow 2} \frac{3x^2 - 5x - 2}{x - 2}$ 1
- e) Find the gradient of the curve $y = 2x^3 - 8x - 1$ at the point $(2, -1)$ 2
- f) Find the coordinates of the point that divides the interval from $(-1, 6)$ to $(4, 3)$ externally in the ratio $5 : 2$. 2

Question 2 (start a new page)

Marks

- a) Find the acute angle between the lines $2x - 4y - 3 = 0$ and $y = 3x - 4$. 2
- b) For what values of k does the equation $(k + 3)x^2 - kx + 1 = 0$ have
i) equal roots 2
ii) real roots 1
- c) A circle with centre $(-4, 1)$ has one end of a diameter at $(-10, -4)$. 1
Find the coordinates of the other end of this diameter.
- d) Find the equation of the tangent to $y = \sqrt{4x + 1}$ at the point $(2, 3)$ 3
- e) Express $\frac{\sin x}{1 + \cos x}$ in terms of t where $t = \tan\left(\frac{x}{2}\right)$. 2

Question 3 (start a new page)

- a) Evaluate $\lim_{x \rightarrow \infty} \frac{8 - 2x^3}{2 + x^2 + x^3}$ 1
- b) Show that $\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} = 4 \cos 2\theta$ 3
- c) The line $ax + by + 3 = 0$ is parallel to $3x + 2y - 4 = 0$ 3
and passes through the point $(1, -2)$. Find the values of a and b .
- d) i) Find the exact value of $\tan 75^\circ$ 2
ii) Use the above result to show that $\tan 75^\circ + \cot 75^\circ = 4$ 2

Question 4 (start a new page)

Marks

a) Differentiate the following with respect to x :

i) $y = \frac{1}{x^2}$ 1

ii) $y = \frac{4x^2}{x+1}$ 2

iii) $y = (x^2 - 1)(x+1)^4$ 2

b) Solve $\sin \theta = \sin 2\theta$ for $0^\circ \leq \theta \leq 360^\circ$ 3

c) Find the equation of the line which passes through the point of intersection of $x - 4y + 5 = 0$ and $2x - 3y - 1 = 0$ and the point $(2,3)$, giving your answer in general form. 3

Question 5 (start a new page)

a) The line $y = mx - 4$ is a tangent to the curve $y = x^2 - 8x$. 3

Find the possible values of m .

b) A and B have coordinates $(-1,7)$ and $(5,-2)$ respectively. 3

P divides the interval AB internally in the ratio $k:1$.

Find the value of k given that P lies on the line $5x - 4y - 1 = 0$

c) Simplify $\frac{\sin 2\alpha}{1 - \cos 2\alpha}$ 2

d) Solve the equation $4 \sin \theta - 2 \cos \theta = 3$ 3

for $0^\circ \leq \theta \leq 360^\circ$ giving answer correct to the nearest degree.

EXT 1 SOLUTIONS JULY 2007

Q1

a) max when $x = -2$

$\therefore \text{max} = 12$

b) $2x^2 - 9x - 11 < 0$

$(2x - 11)(x + 1) < 0$

$-1 < x < 5\frac{1}{2}$

c) i) $\cos(90^\circ - A)$

$= \sin A$

$= \frac{1}{3}$

ii) $\sin 2A$

$= 2 \sin A \cos A$

$= 2 \times \frac{1}{3} \times \frac{\sqrt{8}}{3}$

$= \frac{4\sqrt{2}}{9}$

d) $\lim_{x \rightarrow 2} \frac{(3x+1)(x-2)}{x-2}$

$= 7$

e) $y' = 6x^2 - 8$

when $x = 2$

$m = 6 \times 2^2 - 8$

$= 16$

f) $\left(\frac{20+2}{3}, \frac{15-12}{3} \right)$

$= \left(\frac{22}{3}, 1 \right)$

Q2

a) $m_1 = \frac{1}{2} \quad m_2 = 3$

$\tan \theta = \left| \frac{3 - \frac{1}{2}}{1 + 3 \times \frac{1}{2}} \right|$

$= 1$

$\therefore \theta = 45^\circ$

b) i) $\Delta = 0$

$k^2 - 4(k+3)(1) = 0$

$k^2 - 4k - 12 = 0$

$(k-6)(k+2) = 0$

$k = 6, -2$

ii) $\Delta \geq 0$

$k \leq -2, k \geq 6$

c) $\frac{-10+x}{2} = -4 \quad \frac{-4+y}{2} = 1$

$x = 2$

$y = 6$

$\therefore (2, 6)$

d) $y' = \frac{1}{2} (4x+1)^{-\frac{1}{2}} \cdot 4$

when $x = 2$

$y' = \frac{2}{3}$

$\therefore y - 3 = \frac{2}{3}(x - 2)$

$2x - 3y + 5 = 0$

e) $\frac{2t}{1+t^2} \div \left(1 + \frac{1-t^2}{1+t^2} \right)$

$= \frac{2t}{1+t^2} \times \frac{1+t^2}{1+t^2+1-t^2}$

Q3

a) -2

$$\begin{aligned}
 \text{b) LHS} &= \frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} \\
 &= \frac{\cos \theta \sin 3\theta + \sin \theta \cos 3\theta}{\sin \theta \cos \theta} \\
 &= \frac{2 \sin 4\theta}{\sin 2\theta} \\
 &= \frac{4 \sin 2\theta \cos 2\theta}{\sin 2\theta} \\
 &= 4 \cos 2\theta \\
 &= \text{RHS}
 \end{aligned}$$

c) $-\frac{a}{b} = -\frac{3}{2}$
 $\therefore 2a = 3b \quad (1)$

sub (1, -2)

$$a - 2b + 3 = 0$$

$$a - 2b = -3 \quad (2)$$

Solve (1), (2) simultaneously

$$a = 9, \quad b = 6$$

d) i) $\tan 75^\circ = \tan (45^\circ + 30^\circ)$
 $= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$
 $= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$
 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$

ii) $\tan 75^\circ + \cot 75^\circ$
 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} + \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$
 $= \frac{(\sqrt{3} + 1)^2 + (\sqrt{3} - 1)^2}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$
 $= \frac{8}{2}$
 $= 4$

Q4

a) i) $y' = -2x^{-3}$

ii) $y' = \frac{(x+1)8x - 4x^2(1)}{(x+1)^2}$
 $= \frac{4x^2 + 8x}{(x+1)^2}$

iii) $y' = 2x(x+1)^4$
 $+ (x^2 - 1)4(x+1)^3$
 $= 2(x+1)^3(3x^2 + x - 2)$

b) $\sin \theta = \sin 2\theta$
 $\sin \theta - 2\sin \theta \cos \theta = 0$
 $\sin \theta (1 - 2\cos \theta) = 0$
 $\theta = 0^\circ, 180^\circ, 360^\circ, 60^\circ, 300^\circ$

c) $x - 4y + 5 + k(2x - 3y - 1) = 0$
sub (2, 3)
 $k = -\frac{5}{6}$

Q5

a) $y = mx - 4$
 $y = x^2 - 8x$
 Solve simultaneously
 $x^2 - 8x = mx - 4$
 $x^2 - (m+8)x + 4 = 0$

tangent $\Rightarrow$ 1 solution
 $\Rightarrow \Delta = 0$

$$(m+8)^2 - 4 \times 1 \times 4 = 0$$

$$m^2 + 16m + 48 = 0$$

$$(m+12)(m+4) = 0$$

$$m = -12, -4$$

b) $P \left(\frac{1x-1+kx5}{k+1}, \frac{1x7+kx-2}{k+1} \right)$

$$= \left(\frac{5k-1}{k+1}, \frac{7-2k}{k+1} \right)$$

sub. P into $5x - 4y - 1 = 0$

$$5 \left(\frac{5k-1}{k+1} \right) - 4 \left(\frac{7-2k}{k+1} \right) - 1 = 0$$

$$32k = 34$$

$$k = \frac{17}{16}$$

c) $\frac{\sin 2\alpha}{1 - \cos 2\alpha} = \frac{2 \sin \alpha \cos \alpha}{1 - (1 - 2 \sin^2 \alpha)}$

$$= \frac{\cos \alpha}{\sin \alpha}$$

$$= \cot \alpha$$

d) $4 \sin \theta - 2 \cos \theta = 3$
 $R \sin(\theta - \alpha) = 3$

$$R = \sqrt{20} \quad \tan \alpha = \frac{1}{2}$$

$$\alpha = 26.57^\circ$$

$$\therefore \sqrt{20} \sin(\theta - \alpha) = 3$$

$$\sin(\theta - \alpha) = \frac{3}{\sqrt{20}}$$

$$\theta - \alpha = 42.13^\circ, 137.87^\circ$$

$$\theta = 69^\circ, 164^\circ$$